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Modified atmosphere packaging in the export chain of tropical fruits: efficiency, limitations, and logistical implications

Embalagem de atmosfera modificada na cadeia de exportação de frutas tropicais: eficiência, limitações e implicações logísticas

Envases de atmósfera modificada en la cadena de exportación de frutas tropicales: eficiencia, limitaciones e implicaciones logísticas

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ABSTRACT
Modified Atmosphere Packaging (MAP) has been consolidated as a strategic technology in the postharvest conservation of tropical fruits, contributing to the reduction of losses, the maintenance of quality, and the feasibility of exports to distant markets. This article presents a critical review of the advances, limitations, and logistical implications of MAP, with emphasis on mangoes, based on the analysis of national and international scientific literature published between 2010 and 2025, complemented by consolidated studies from previous decades. The review highlights the physiological principles of MAP, the distinctions between passive and active approaches, the importance of proper packaging material selection, and its interaction with refrigeration. Practical cases with mango cultivars such as 'Keitt', Ataulfo', Tommy Atkins', and 'Palmer' demonstrate that the response to the technique depends on the cultivar, film permeability, and logistical conditions. Although complementary technologies such as ethylene filters may add value in certain contexts, the data indicate that their application should be carefully evaluated. The results suggest that the effectiveness of MAP is directly related to technical planning, understanding of product physiology, and the adaptation of solutions to the operational realities of the export chain. It is concluded that, when properly applied, MAP can represent a competitive advantage for tropical fruit production, expanding access to international markets with lower environmental impact and greater added value.

Keywords: Ethylene, MAP, O_2, CO_2, Postharvest, Conservation.

RESUMO
A embalagem de atmosfera modificada (AM) tem se consolidado como uma tecnologia estratégica na conservação pós-colheita de frutas tropicais, contribuindo para a redução

de perdas, a manutenção da qualidade e a viabilização da exportação para mercados distantes. Este artigo apresenta uma revisão crítica sobre os avanços, limitações e implicações logísticas da AM, com ênfase em mangas, a partir da análise de literatura científica nacional e internacional publicada entre 2010 e 2025, complementada por estudos consolidados das décadas anteriores. A revisão destaca os princípios fisiológicos da AM, as distinções entre abordagens passiva e ativa, a importância da seleção adequada de materiais de embalagem e sua interação com a refrigeração. Casos práticos com cultivares de manga como 'Keitt', 'Ataulfo', 'Tommy Atkins' e 'Palmer' demonstram que a resposta à técnica é dependente da cultivar, da permeabilidade do filme e das condições logísticas. Embora tecnologias complementares como filtros de etileno possam agregar valor em alguns contextos, os dados indicam que sua aplicação deve ser criteriosa. Os resultados sugerem que a eficácia da AM está diretamente relacionada ao planejamento técnico, à compreensão da fisiologia do produto e à adaptação das soluções às realidades operacionais da cadeia de exportação. Conclui-se que a AM, quando corretamente aplicada, pode representar um diferencial competitivo para a fruticultura tropical, ampliando o acesso a mercados internacionais com menor impacto ambiental e maior valor agregado.

Palavras-chave: Etileno, AM, O_2, CO_2, Pós-Colheita, Conservação.

RESUMEN

El envasado en atmósfera modificada (AM) se ha consolidado como una tecnología estratégica en la conservación poscosecha de frutas tropicales, contribuyendo a la reducción de pérdidas, al mantenimiento de la calidad y a la viabilidad de la exportación hacia mercados distantes. Este artículo presenta una revisión crítica sobre los avances, limitaciones e implicaciones logísticas de la AM, con énfasis en mangos, a partir del análisis de literatura científica nacional e internacional publicada entre 2010 y 2025, complementada con estudios consolidados de décadas anteriores. La revisión destaca los principios fisiológicos de la AM, las diferencias entre los enfoques pasivo y activo, la importancia de la selección adecuada de materiales de envasado y su interacción con la refrigeración. Casos prácticos con cultivares de mango como 'Keitt', 'Ataulfo', 'Tommy Atkins' y 'Palmer' demuestran que la respuesta a la técnica depende de la variedad, de la permeabilidad del film y de las condiciones logísticas. Aunque tecnologías complementarias como los filtros de etileno pueden aportar valor en algunos contextos, los datos indican que su aplicación debe ser criteriosa. Los resultados sugieren que la eficacia de la AM está directamente relacionada con la planificación técnica, la comprensión de la fisiología del producto y la adaptación de las soluciones a las realidades operativas de la cadena de exportación. Se concluye que la AM, cuando se aplica correctamente, puede representar una ventaja competitiva para la fruticultura tropical, ampliando el acceso a los mercados internacionales con menor impacto ambiental y mayor valor agregado.

Palabras clave: Etileno, AM, O_2, CO_2, Poscosecha, Conservación.

1 INTRODUCTION

The postharvest conservation of fruits and vegetables represents one of the greatest challenges for contemporary agriculture due to their high perishability and the significant losses that occur throughout the supply chain. It is estimated that more than 30% of global fruit and vegetable production is lost between harvest and consumption, resulting in substantial economic, social, and environmental impacts (Oyewole *et al*., 2025). These losses are mainly attributed to the intense metabolic activity of fresh produce and their susceptibility to mechanical and microbiological damage after harvest.

To reduce such losses and extend shelf life, a variety of preservation technologies have been developed and applied, including refrigeration, chemical treatments, and edible coatings. Among these, Modified Atmosphere Packaging (MAP) stands out as a practical, efficient, and economically viable strategy for the conservation of fresh fruits and vegetables (Harish, 2024; Mikus & Galus, 2025). This technique consists of modifying the gas composition surrounding the product, usually by reducing oxygen (O_2) levels and increasing carbon dioxide (CO_2), which lowers respiration rate, decreases ethylene production, and delays ripening and tissue senescence (Moradi & Ahmadi, 2024; Romansyah *et al*., 2025).

The concept of modifying the environment around food is not new: in Ancient Egypt, foods were already stored in hermetically sealed containers. In 1821, Jacquet Beard conducted the first fruit preservation experiments in France; however, it was only in the early 20th century that Kidd and West, in England, systematized the technique that would later evolve into Controlled Atmosphere Storage (CA) (Brackmann, 2006). The term “modified atmosphere” was introduced to distinguish this technique, where the final gas composition is not rigidly controlled but results from the balance between product respiration and film permeability, from traditional controlled atmosphere methods (Kader, 1992).

MAP can be achieved passively, when product respiration and film properties interact until a new atmospheric equilibrium is reached, or actively, through the deliberate introduction of a specific gas mixture before sealing (Pooja & Kaushal, 2024). In both

cases, careful selection of packaging material and knowledge of the physiological characteristics of the product are essential for the success of the technique.

Beyond O_2 and CO_2, other gases may be used in MAP, such as nitrogen (N_2), which is chemically inert and serves as a filler gas to reduce oxidation and prevent package collapse (Church, 1993). CO_2 also exhibits antimicrobial effects, more pronounced at low temperatures, and competes with ethylene, thus delaying ripening (Franco, 2004). Meanwhile, reduced O_2 concentrations suppress respiratory activity and microbial proliferation (Sandhya, 2010).

The effectiveness of MAP has been demonstrated across a wide range of commodities. In bell peppers, Romansyah *et al*. (2025) reported that MAP reduced weight loss, maintained color, and increased firmness during storage. Ju *et al*. (2025) obtained similar results for Korean melons, emphasizing their export potential. For apples, Mikus and Galus (2025) reviewed the benefits of edible films associated with MAP, documenting reduced respiration and lower pathogen incidence, while Sachin *et al*. (2025) showed that passive MAP extended guava shelf life by up to 21 days under low temperatures.

Other studies highlight the constructive collaboration of MAP with additional technologies, such as plant-based edible coatings (Champa & Weerasooriya, 2025), antimicrobial nanofilms for cherries (Zhuang *et al*., 2025), and intelligent films capable of responding to internal conditions (Maryam *et al*., 2025). Despite promising results, many articles and manuals still address MAP only superficially, often as a subchapter in broader postharvest texts, without exploring the most recent technological advances (Harish, 2024; Mikus & Galus, 2025). Furthermore, challenges such as the selection of sustainable materials, consumer acceptance, and compliance with sanitary regulations remain critical issues that require clarification (Maryam *et al*., 2025; Oyewole *et al*., 2025).

This study was designed to bridge these gaps by providing a detailed and up-to-date analysis of MAP, covering its physiological and physicochemical foundations, packaging materials and techniques, practical applications, and emerging trends. The aim is to provide researchers, students, and industry professionals with a solid basis to

understand, apply, and innovate in the use of MAP for the conservation of fruits and vegetables in an efficient and sustainable manner.

2 LITERATURE REVIEW

2.1 FUNDAMENTALS OF MODIFIED ATMOSPHERE PACKAGING

2.1.1 Basic Concepts and Physiological Principles

Postharvest quality of fruits and vegetables is directly related to the intensity of physiological and biochemical processes that continue after harvest. The most relevant of these processes is respiration, which involves the consumption of oxygen (O_2) and the release of carbon dioxide (CO_2), heat, and water. The higher the respiration rate, the faster the degradation of tissues, leading to accelerated ripening, weight loss, softening, color changes, and increased susceptibility to pathogens (Siddiq *et al*., 2020).

Modified Atmosphere Packaging (MAP) was developed to intervene directly in this process, creating an internal atmosphere less favorable to metabolic activity. The basic principle consists of sealing the fresh product in a polymeric film with specific gas permeability, so that product respiration and film properties interact to gradually reduce O_2 levels and increase CO_2. This microenvironment delays ripening, inhibits ethylene production, and reduces the incidence of diseases, thereby extending shelf life and maintaining product quality (Altieri *et al*., 2018; Siddiq *et al*., 2020; Vilvert *et al*., 2023).

2.1.2 Controlled vs. Modified Atmosphere

It is important to distinguish between Controlled Atmosphere (CA) and Modified Atmosphere (MAP). In both systems, the composition of gases surrounding the product is altered, with reduced O_2 and increased CO_2. However, in CA, gas concentrations are adjusted and maintained precisely throughout storage through continuous injection and monitoring. In MAP, there is no precise control: the modification occurs naturally and

gradually because of product respiration combined with film permeability. This process is highly dependent on the product's respiration rate, storage temperature, package surface area, and film diffusion properties (Freitas *et al*., 2023; Mongollón *et al*., 2024).

For products such as strawberries, studies indicate that target atmospheres of 3–5% O_2 and 10–15% CO_2 at 1 °C promote significant preservation of firmness, color, and reduction of decay (Mongollón *et al*., 2024; Silva *et al*., 2023). The combined use of MAP and antimicrobial sachets has shown superior performance compared to MAP alone, extending shelf life and enhancing microbiological safety.

In 'Niagara Rosada' grapes, the use of low-density polyethylene (LDPE) film under passive MAP reduced weight loss but was insufficient to control berry drop and fungal diseases. Ethylene-absorbing sachets, in this case, also showed no additional benefits (Auri *et al*., 2024). In 'Tommy Atkins' mangoes, the lower physiological limit for O_2 was identified at 2%; below this level, fermentation and tissue breakdown occurred, even with BreatheWay® membranes and storage at 10 °C (Freitas *et al*., 2023). This demonstrates that MAP has critical limits that cannot be exceeded without physiological risks.

For 'Fuyu' persimmons, two independent studies reinforced the value of MAP. The first (Auri *et al*., 2024) showed that packages with medium gas barrier properties (reaching ~6% O_2 and 12% CO_2) provided better maintenance of firmness and external appearance. The second (Silva *et al*., 2023) observed that packages with lower permeability, when combined with refrigeration, prevented internal browning and extended shelf life.

In 'Rocha' pears, the application of dynamic controlled atmosphere, with real-time monitoring of fruit respiration, was effective in preventing internal browning, a common disorder under very low O_2 conditions. The advantage of CA in this case was its ability to automatically adjust to physiological limits, thereby avoiding anaerobic damage (Rodrigues *et al*., 2024). These findings reinforce that while MAP offers a practical and cost-effective alternative, especially for simpler logistics chains, CA is preferable in cases requiring precision, such as fruits with narrow O_2 tolerance thresholds. The choice between CA and MAP must consider the species' physiology, storage duration, temperature, operational costs, and the commercial destination of the product.

2.1.3 Active and Passive MAP

Modified Atmosphere Packaging (MAP) can be achieved through two main approaches:

a) **Passive MAP:** The atmospheric modification occurs exclusively through product respiration and gas diffusion across the packaging material. It is the simplest and most economical method, although it requires greater caution to avoid hypoxia (Kader & Watkins, 2000).

b) **Active MAP:** A specific gas mixture is introduced at the moment of sealing, accelerating the establishment of the desired atmosphere and allowing greater initial control (Rico *et al*., 2007; Mangaraj *et al*., 2009).

2.1.4 Role of Film Permeability

The permeability of plastic films is a critical factor for MAP performance. Films with very low permeability to O_2 and CO_2 can induce anoxia and fermentation if not properly designed. Conversely, highly permeable films may fail to achieve the concentrations necessary to slow physiological processes. Therefore, the choice of film must take into account the product's respiration rate, sensitivity to CO_2, storage temperature and humidity, and the desired shelf life. Understanding these principles is essential for the correct application of the technique. Errors in package design or in the choice between passive and active MAP may compromise final product quality.

2.1.5 Interaction with Refrigeration

Postharvest of fruits and vegetables is largely determined by physiological processes such as respiration, which continue even after harvest. During respiration, O_2 is consumed, while CO_2, water, and heat are released. The more intense the respiration, the faster the tissue degradation, leading to accelerated ripening, loss of firmness, and greater susceptibility to pathogens. Refrigeration reduces the respiration rate and prolongs shelf

life by lowering water vapor pressure within tissues, thereby preventing evaporation and loss of turgor, which helps maintain sensory quality (Chitarra & Chitarra, 2005). However, refrigeration alone is often insufficient to contain all physical and microbiological deterioration. For this reason, its association with MAP has been widely studied as a complementary strategy.

According to Brackmann *et al*. (2004), MAP is an alternative that enhances the effect of cold storage. When used together with refrigeration, MAP or controlled atmosphere can reduce respiration rates by up to 50% compared to refrigeration alone (Chitarra & Chitarra, 2005). Modified atmosphere storage allows for longer preservation because it combines high relative humidity with controlled partial pressures of O_2 and CO_2 inside the package, thus decreasing metabolic activity (Chitarra & Chitarra, 2005). The rate at which the internal atmosphere is modified depends on the respiration rate of the product, storage temperature, humidity, and film permeability (Sarantópoulos *et al*., 1996; Cia, 2002). According to Parry (1993), MAP can extend shelf life by three to four times compared to refrigeration alone, meeting consumer demand for fresh, high-quality, preservative-free products.

2.1.6 Practical Applications

Studies on different fruits have shown both the benefits and the limitations of combining MAP with refrigeration. For blackberries (Guarani' and 'Caingangue' cultivars) stored at 5 °C and 90% relative humidity, the use of LDPE films under MAP significantly reduced weight loss and preserved soluble solids, acidity, and pH compared to unpackaged fruits. However, retained moisture inside the package favored decay, especially in the 'Guarani' cultivar, indicating the need for film adjustments to improve ventilation (Cia *et al*., 2007).

In 'Niagara Rosada' grapes stored for 28 days at 1 °C and 90% relative humidity, MAP was also effective in reducing weight loss and maintaining visual appearance. The best results were obtained with twenty-five µm LDPE films, which provided a good bal-

ance between permeability and moisture retention. However, the technique did not adequately control decay incidence or berry drop, highlighting the importance of carefully selecting materials and storage conditions (Cia *et al*., 2010).

For mangoes, results are also positive but heterogeneous across cultivars and packaging types. Pfaffenbach *et al*. (2003) tested 'Espada Vermelha' mangoes at 12 °C with different materials and found that LDPE combined with ethylene-absorbing sachets preserved quality for up to 28 days under refrigeration with immediate consumption, or 14 days followed by 4 days at ambient temperature. PVC films also reduced weight loss, though they were less effective than LDPE combined with sachets. In contrast, Conservax negatively affected fruit coloration and ripening, demonstrating that packaging material directly influences sensory attributes.

In a more recent study, Mello Júnior (2025) evaluated 'Ataulfo' and 'Keitt' mangoes under refrigeration combined with MAP and confirmed the technique's effectiveness in extending shelf life and preserving fruit quality. MAP significantly reduced weight loss and respiration rates, maintaining color and firmness for longer periods. This study underscores the importance of considering varietal characteristics and adjusting film properties and storage conditions to maximize benefits while minimizing risks such as anaerobiosis or pathogen growth.

2.1.7 Packaging Materials and Technologies

The appropriate selection of packaging materials and technologies is essential for the success of postharvest conservation of fruits and vegetables. Packaging not only protects products against mechanical damage and contamination but also acts as a functional barrier that regulates the internal atmosphere, controls humidity, and balances transpiration. These functions are particularly critical for commodities with high respiratory rates, in which an inadequate environment can accelerate fermentation, wilting, or pathogen development (Ribeiro *et al*., 2005; Fang *et al*., 2021).

The main functions of packaging include physical protection against shocks and compression during transportation and handling; a microbiological barrier that reduces

exposure to external contaminants; moisture regulation, essential to prevent water loss and dehydration; and the control of the internal atmosphere, which allows adjustment of oxygen (O_2) and carbon dioxide (CO_2) concentrations, promoting conservation through reduced respiration and delayed ripening (Batista *et al*., 2007; Brackmann *et al*., 1999).

The materials used for packaging must have permeability properties compatible with the product's respiration rate. Low-density polyethylene (LDPE), widely used, is notable for its flexibility and low water vapor permeability, making it suitable for commodities with low to moderate respiration rates, although it may favor internal condensation (Finger & Vieira, 1997). High-density polyethylene (HDPE) offers greater rigidity and a stronger barrier, making it more suitable for products less sensitive to CO_2 accumulation (Batista *et al*., 2007). Polypropylene (PP) provides high transparency, mechanical strength, and intermediate permeability, often used in biaxially oriented films (BOPP) with potential to create protective atmospheres in more sensitive products (Brackmann *et al*., 1999). Polyvinyl chloride (PVC), in turn, exhibits greater permeability to gases and water vapor, reducing condensation risk in high-respiration products (Batista *et al*., 2007). Multilayer and composite films allow for selective barrier combinations and may include antimicrobial layers or ethylene scavengers (Wang *et al*., 2022).

Regarding packaging technologies, the modified atmosphere can be established either passively or actively. In passive MAP, atmospheric modification occurs gradually after sealing, because of the balance between product respiration and film permeability (Masterpack, 2020). This approach is simpler and more economical but less precise. In active MAP, a specific gas mixture is inserted now of sealing, allowing the immediate attainment of desired O_2 and CO_2 concentrations and providing greater control over the initial microenvironment (Fang *et al*., 2021).

Complementary technologies may also be incorporated, such as ethylene, oxygen, or moisture absorbers, which can be included as sachets or integrated into the material. These devices are particularly useful for climacteric fruits, sensitive to ethylene action (Ethylene absorber, 2023; Fang *et al*., 2021). Intelligent packaging systems incorporate sensors or visual indicators capable of monitoring variables such as temperature, gas con-

centrations, or volatile compounds, signaling changes that may compromise product quality. Examples include colorimetric indicators that change color when critical levels of ethylene or improper temperatures are detected (Huang *et al*., 2025).

Recent innovations include microperforated films, which contain small, uniformly distributed perforations and are ideal for high-respiration products. This technology improves gas exchange and reduces condensation risk (Microperforated Packaging, 2023). Another relevant innovation is antimicrobial films, containing active agents capable of inhibiting fungal and bacterial growth, either incorporated into the polymer matrix or applied on the surface, thus prolonging safety and shelf life (Trends in Food Science & Technology, 2020).

Sustainability has gained prominence in recent years. Strategies such as the use of biodegradable biopolymers (based on PLA, starch, or cellulose), reduced film thickness, recyclable materials, and reusable packaging have been adopted to minimize environmental impacts (Hongren, 2021). However, many of these sustainable materials still show inferior performance compared to conventional polymers, particularly regarding gas barrier properties and mechanical resistance, which requires further research and development to enhance their properties and enable large-scale commercial application (Wang *et al*., 2022).

2.2 PRACTICAL CONSIDERATIONS

The choice of packaging material and technology must be made carefully, considering the respiration rate of the product, its sensitivity to carbon dioxide and ethylene, as well as the expected transportation and storage conditions. In addition to these factors, it is essential to consider the desired shelf life and evaluate the cost–benefit balance between using simpler materials and adopting advanced technologies. To ensure that these parameters are properly adjusted, preliminary tests are recommended, since errors in package design may lead to undesirable situations, such as anaerobiosis or excessive moisture accumulation, compromising final product quality.

2.2.1 Applications in Fruits

Postharvest conservation of fruits represents one of the greatest challenges in the supply chain, due to their high perishability and the intense physiological transformations that occur after harvest. Fruits are living, metabolically active structures, with high respiration and transpiration rates, in addition to being highly susceptible to pathogen attack. As a result, their shelf life is limited, requiring preservation techniques capable of delaying ripening, reducing water loss, and minimizing disease development.

Among the available technologies, the combination of refrigeration with Modified Atmosphere Packaging (MAP) has emerged as one of the most promising strategies. This approach allows for adjustment of the microenvironment around fruits, controlling oxygen and carbon dioxide concentrations to reduce respiration and preserve quality attributes such as color, firmness, flavor, and nutritional value.

This section presents case studies and practical results of MAP applications in different fruits. Examples with specific cultivars illustrate physiological responses, effects on postharvest quality, and the limitations observed in each situation. The objective is to provide a comprehensive overview of the applications of this technology in fruits, serving as a reference for professionals and researchers seeking to implement or improve these techniques in practice.

2.2.2 Case Studies with ‘Keitt’, ‘Ataulfo’, ‘Tommy Atkins’, and ‘Palmer’ Mangoes

Among tropical fruits, mango stands out as one of Brazil’s leading export commodities, but also as one of the most challenging from a postharvest perspective, due to its high respiration rate and susceptibility to physiological disorders and diseases. The following studies conducted in the Submédio São Francisco Valley evaluated the effectiveness of MAP and ethylene-absorbing filters in extending shelf life and maintaining quality during transport and commercialization.

2.2.3 'Keitt' and 'Ataulfo' Mangoes

In both 'Keitt' and 'Ataulfo' mangoes, MAP has been shown to effectively reduce weight loss, respiration rate, and physiological disorders while maintaining firmness and soluble solids during refrigerated storage (Mello Júnior, 2025). Low-permeability films generally produced the best results, particularly in 'Keitt', where they significantly extended shelf life and preserved pulp quality. In 'Ataulfo', however, responses varied between seasons: in one harvest, medium-permeability films were more effective, while in another, low-permeability films provided superior performance.

These oscillations illustrate that MAP effectiveness is not determined solely by film properties but also by seasonal factors such as climate conditions, harvest maturity, and handling practices. In both cultivars, the use of ethylene filters did not lead to additional improvements, reinforcing that their practical utility under refrigeration may be limited.

By contrast, Brecht *et al*. (2023) demonstrated that MAP combined with ethylene removal improved firmness retention, delayed sugar and acid degradation, and enhanced sensory quality in 'Keitt' mangoes harvested at advanced maturity. These contrasting findings suggest that the role of complementary technologies such as ethylene scrubbing may depend on cultivar-specific physiology, harvest stage, and storage conditions.

2.2.4 'Tommy Atkins' Mangoes

'Tommy Atkins' is one of Brazil's most exported cultivars, but its rapid ripening limits long-distance shipping. Studies conducted during the winter and summer seasons tested packages with different permeabilities (BWA, BWB, BWC), with and without ethylene filters, and perforated packaging with ethylene absorption (EAE) (Monteiro, 2025). Fruits were stored for 28 days at 9 °C, followed by 4 days at 20 °C. All packaging reduced respiration rate and weight loss, but BWC was the most effective in maintaining quality and minimizing internal physiological disorders. Again, ethylene filters showed no additional significant effect.

Furthermore, Brecht *et al*. (2023) confirmed the feasibility of MAP combined with ethylene removal for 'Tommy Atkins', especially when harvested at advanced maturity. Fruits stored under 6% O_2 exhibited better firmness, slower degradation of sugars and acids, and higher sensory acceptance compared to controls, making this strategy promising for international shipping.

3 METHODOLOGY

This critical review was conducted with the aim of systematically compiling, analyzing, and discussing the advances and limitations of Modified Atmosphere Packaging (MAP) technology applied to the postharvest conservation of tropical fruits. The study followed a qualitative and interpretative approach, grounded in the analysis of national and international scientific literature.

Priority was given to studies published between 2010 and 2025 to reflect the most recent advances in the field. However, classical works and consolidated references from the 1990s and 2000s were also incorporated due to their historical and conceptual importance, particularly with respect to the physiological foundations of MAP, the distinctions between controlled and modified atmospheres, and the first commercial applications of the technique.

The bibliographic search was carried out in databases such as ScienceDirect, Scopus, Web of Science, Google Scholar, and institutional repositories such as Embrapa's ALICE, using descriptors in Portuguese and English, including: *"embalagem de atmosfera modificada"*, *"frutas tropicais"*, *"armazenamento refrigerado"*, *"modified atmosphere packaging (MAP)"*, *"shelf life"*, and *"postharvest mango conservation"*. Only publications with direct thematic relevance, practical applicability, or robust scientific grounding were selected, with priority given to studies presenting experimental data under controlled conditions or realistic simulations of transport, marketing, and storage.

Once compiled, the materials were critically analyzed with focus on the most relevant experimental variables, such as fruit type, cultivar, packaging characteristics, ap-

proach (active or passive), storage time and temperature, use of complementary technologies, and final product quality. The analysis sought to identify patterns, divergences, and gaps among the studies, linking the findings to the physiological requirements of the commodities and the actual conditions of the logistics chain. The evidence was synthesized comparatively and discussed considering operational challenges and emerging technological trends, with the purpose of providing a technical and pragmatic perspective on the potential and limitations of MAP in the context of tropical fruit production.

4 RESULTS AND DISCUSSION

Although Modified Atmosphere Packaging (MAP) is widely recognized for its ability to extend the shelf life of fresh fruits, its effectiveness is strongly influenced by physiological, environmental, and operational factors. In mangoes, for instance, studies in the Submédio São Francisco Valley have shown that responses to MAP vary among cultivars and even between consecutive harvests.

For the cultivars 'Ataulfo' and 'Keitt', seasonal differences led to contrasting outcomes, in some cases, films with medium permeability better preserved firmness and coloration, while in other seasons low-permeability films were more effective in reducing weight loss and minimizing physiological disorders (Mello Júnior, 2025). These variations suggest that MAP performance cannot be attributed solely to film characteristics but is also conditioned by factors such as maturity stage at harvest, climate, and handling practices. Ethylene filters, meanwhile, did not significantly improve results under refrigerated conditions, indicating limited functional relevance in this context.

Similar trends have been reported for 'Tommy Atkins' and 'Palmer', where MAP reduced respiration and weight loss, with low-permeability films showing the best performance, though again without additional benefits from ethylene absorption (Monteiro, 2025). By contrast, Brecht *et al*. (2023) found that the combination of MAP with ethylene scrubbing improved firmness, sugar and acid retention, and sensory acceptance in

'Tommy Atkins', 'Kent', and 'Keitt' mangoes destined for export, highlighting that cultivar-specific responses and storage conditions determine whether complementary technologies are advantageous.

Other fruits show comparable patterns. In blackberries, MAP reduced weight loss but condensation favored decay (Cia *et al*., 2007). In 'Niagara Rosada' grapes, films effectively reduced water loss but failed to prevent berry drop and fungal diseases (Cia *et al*., 2010). Such results illustrate the classic paradox of MAP: while slowing transpiration, it can raise internal humidity to levels that favor microbial growth. The optimal balance therefore depends on film permeability tailored to respiration rate, CO_2 tolerance, and logistical conditions. For example, in 'Espada Vermelha' mangoes, LDPE films with ethylene absorbers extended shelf life but impaired peel coloration when combined with certain commercial products (Pfaffenbach *et al*., 2003).

Overall, while new technologies such as antimicrobial films and intelligent packaging are gaining attention (Maryam *et al*., 2025; Zhuang *et al*., 2025), classical polymers like LDPE and PVC, when properly calibrated, continue to deliver consistent and cost-effective results. Thus, MAP should be viewed not as a fixed protocol but as a flexible tool that requires adjustment according to commodity physiology and supply chain conditions.

5 CONCLUSION

Modified Atmosphere Packaging (MAP) stands out as one of the most relevant technologies for the postharvest conservation of highly perishable tropical fruits such as mango. However, its effectiveness is not universal, since it depends on physiological, environmental, and logistical factors that vary among cultivars and seasons. The inappropriate use of materials, inadequate selection of film permeability, or the adoption of complementary technologies without physiological justification can compromise results. Therefore, generic protocols are insufficient in the face of supply chain complexity, reinforcing the need for adaptive and context-specific approaches.

Rather than relying solely on technological innovations, MAP requires technical planning and applied agronomic intelligence that integrates product physiology, refrigeration, and water control. With such an approach, MAP can move beyond being considered a "promising technology" and become established as a reliable and essential component of postharvest strategies for tropical fruit production, with direct contributions to loss reduction, sustainability, and access to competitive international markets.

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